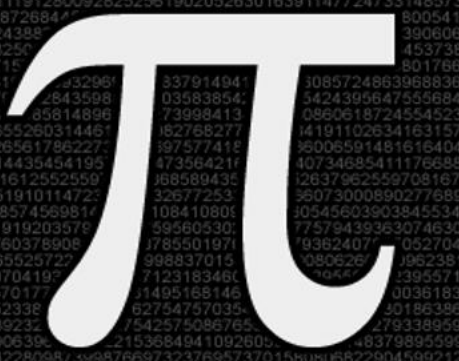


Chapter 2

MATHEMATICAL BACKGROUND – PART 1



Outlines

1. Integer Arithmetic.
2. Modular Arithmetic.
3. Matrices (Optional).

Learning Objectives

In the end of this Chapter 2, students should be able:

1. To perform **integer arithmetic**, including **binary operations**, **integer division**, **greatest common divisor**, and **divisibility**.
2. To perform **modular arithmetic**, including **congruences**, **additive** and **multiplicative inverses**.
3. To review and perform operations on **matrices**, both in integer and modular arithmetic environments.

INTEGER ARITHMETIC



Integers and Binary Operations

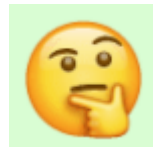
- Set of Integers:

- Denoted by $\mathbb{Z}$, it contains all the **integral numbers** (with **no fraction**), ranging from negative to positive infinities.

$$\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$$

- In cryptography, the interest is in the **three binary operations** (operations that take in two inputs and produce one output) that applied to the set of integers:

1. **Addition**
2. **Subtraction**
3. **Multiplication**



Why Division is **NOT**
part of the binary
operations?

Integer Division

- **Division operation** in general produces **two outputs**, that are quotient q , and remainder r . We define the **division relation** as follows. Given two integers a and n , with $a > n$, then we can express them as:

$$a = q \cdot n + r$$

- In the case of cryptography application, we impose two restrictions:
 1. The integer $n > 0$, that is, it must be **positive**.
 2. The remainder $0 \leq r < n$, that is, it must be **non-negative**.
- If the result contains negative outputs, we simply '**trade in**' some rounds of **quotient q** with **remainder r** to return it into positive.

Divisibility

- From **division relation** $a = q \cdot n + r$, if the remainder $r = 0$, then we have:

$$a = q \cdot n$$

and we say that **n divides a** , or mathematically $n \mid a$. Otherwise, $n \nmid a$.

- Examples:

1. $32 = 8 \times 4$. Hence, $4 \mid 32$, or $8 \mid 32$.

2. $38 = 9 \times 4 + 2$. Hence, $4 \nmid 38$, or $9 \nmid 38$.

Divisibility – Properties

- The following outline several **properties of divisibility**.
 1. If $a \mid 1$, then $a = \pm 1$.
 2. If $a \mid b$, and $b \mid a$, then $a = \pm b$.
 3. If $a \mid b$, and $b \mid c$, then $a \mid c$.
 4. If $a \mid b$, and $a \mid c$, then $a \mid (m \cdot b + n \cdot c)$, where m, n are arbitrary integers.
- ❖ You are **not required** to understand the proof behind these properties, but you may look into it if you are interested to know '**WHY?**'.

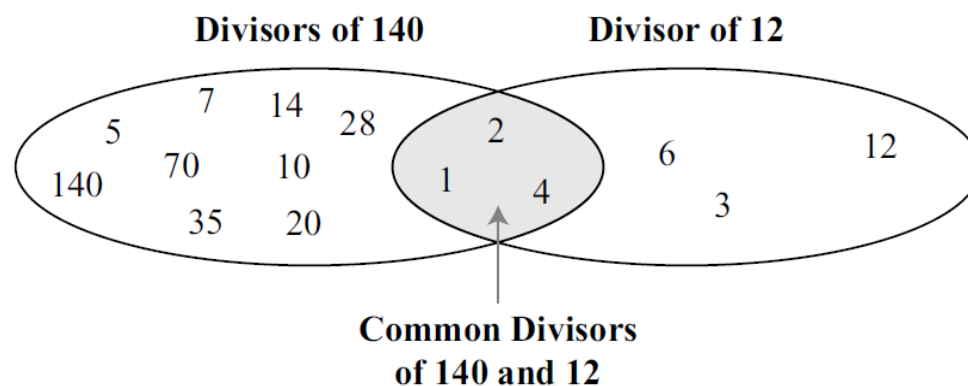
Divisibility – Divisors

- A **divisor** is an integer that can divide another integer, either **completely** or with a **remainder**. In this course, we consider only the case of ‘**completely**’.
- In this case, we observe the two interesting facts:
 1. The integer 1 has only one divisor, which is itself.
 2. ANY positive integer has **at least two divisors**, which are 1 and itself.
- Examples:
 1. 7 has only two divisors, which are 1, 7. (You will soon learn about the Prime Numbers)
 2. 32 has six divisors, which are 1, 2, 4, 8, 16, 32.

Divisibility – Greatest Common Divisor

- In cryptography, the one that we are needed is the **greatest common divisor**, or simply gcd of **two (or more) positive integers**.
- Any two integers may have many common divisors, but only one **GREATEST (largest)** common divisor.

Figure 2.6 *Common divisors of two integers*



The common divisors of 12 and 140 are 1, 2, 4.

The **Greatest** Common Divisor of 12 and 140 is 4.

Divisibility – Greatest Common Divisor

- Definition 1:

The greatest common divisor of two integers is the largest integer that can divide both integers.

- As in the previous slide, using mathematical notation, we write:

$$\mathbf{gcd(12, 140) = 4.}$$

- **Important case in Cryptography:** If the $\mathbf{gcd(a, b) = 1}$, then we say that the integers a and b are **relatively prime (coprime)**.

Divisibility – The Euclidean Algorithm

- The Euclidean algorithm:

- An efficient algorithm used to find the **gcd** of two integers (since listing out all the divisors is **impractical** for a very large integer. Example: 65525).

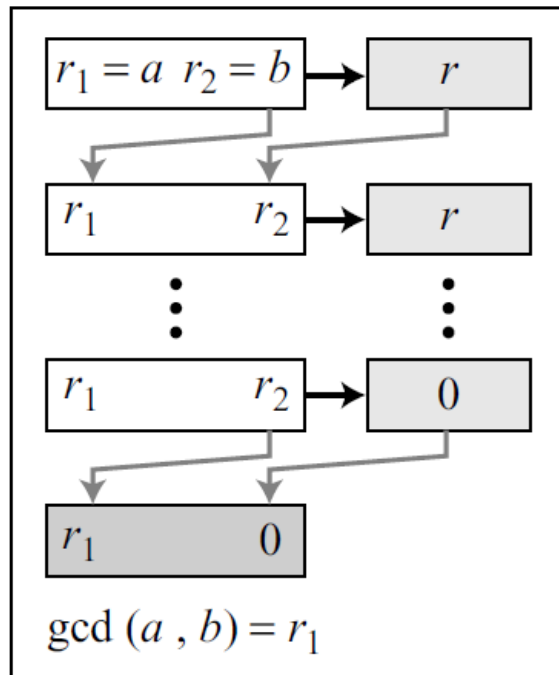
- This algorithm is based on these **two facts**:

- 1. $\gcd(a, 0) = a$.

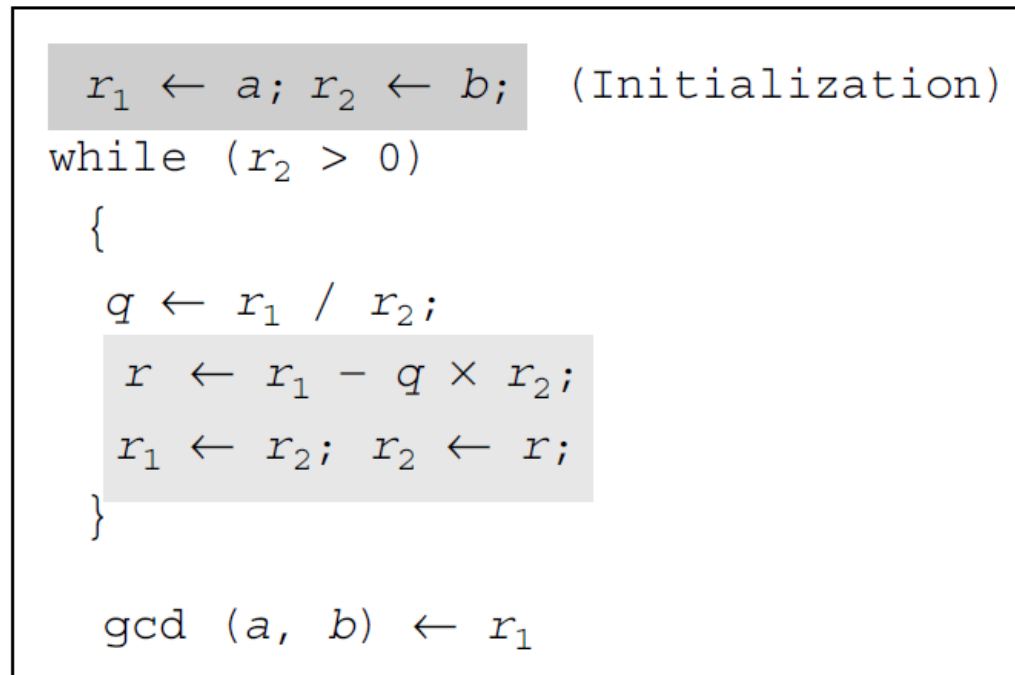
- 2. $\gcd(a, b) = \gcd(b, r)$, where r is the remainder of dividing a by b .

Divisibility – The Euclidean Algorithm

Figure 2.7 *Euclidean algorithm*



a. Process



b. Algorithm

- **Example 1:** Find the gcd of 2740 and 1760.

[illegible]

Divisibility – The Euclidean Algorithm

- **Example 1:** Find the gcd of 2740 and 1760.

q	r_1	r_2	r
1	2740	1760	980
1	1760	980	780
1	980	780	200
3	780	200	180
1	200	180	20
9	180	20	0
	20	0	

- Hence, $\text{gcd}(2740, 1760) = \mathbf{20}$.

- **Example 2:** Find the gcd of 25 and 60.

[illegible]

Divisibility – The Euclidean Algorithm

- **Example 2:** Find the gcd of 25 and 60.

q	r_1	r_2	r
0	25	60	25
2	60	25	10
2	25	10	5
2	10	5	0
	5	0	

- Hence, $\gcd(25,60) = \mathbf{5}$.

Divisibility – The Extended Euclidean Algorithm

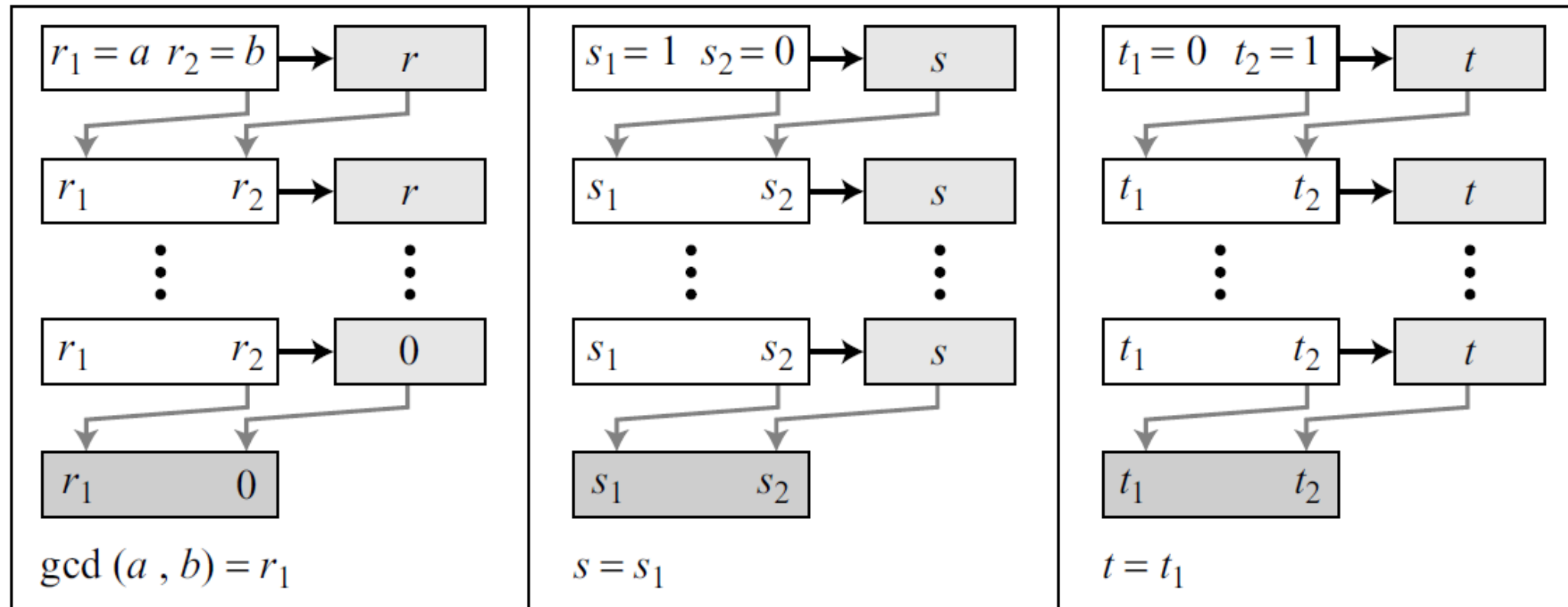
- Definition 2:

Given two integers a and b , there exist another two integers s and t such that

$$s \cdot a + t \cdot b = \gcd(a, b)$$

- These integers s and t can be found via the Extended Euclidean Algorithm.
- It is so-called ‘extended’ as we used the Euclidean algorithm to further finding these two integers, with some extended computations.

Divisibility – The Extended Euclidean Algorithm



```
 $r_1 \leftarrow a; r_2 \leftarrow b;$   
 $s_1 \leftarrow 1; s_2 \leftarrow 0;$  (Initialization)  
 $t_1 \leftarrow 0; t_2 \leftarrow 1;$ 
```

```
while ( $r_2 > 0$ )
```

```
{
```

```
   $q \leftarrow r_1 / r_2;$ 
```

```
     $r \leftarrow r_1 - q \times r_2;$   
     $r_1 \leftarrow r_2; r_2 \leftarrow r;$  (Updating  $r'$ s)
```

```
     $s \leftarrow s_1 - q \times s_2;$   
     $s_1 \leftarrow s_2; s_2 \leftarrow s;$  (Updating  $s'$ s)
```

```
     $t \leftarrow t_1 - q \times t_2;$   
     $t_1 \leftarrow t_2; t_2 \leftarrow t;$  (Updating  $t'$ s)
```

```
}
```

```
gcd ( $a, b$ )  $\leftarrow r_1; s \leftarrow s_1; t \leftarrow t_1$ 
```

- **Example 1:** Find the $\gcd(161, 28)$. Next, find the values of s and t such that

$$161 \cdot s + 28 \cdot t = \gcd(161, 28)$$

[illegible]

Divisibility – The Extended Euclidean Algorithm

- Example 1: Find the $\gcd(161, 28)$. Next, find the values of s and t such that

$$161 \cdot s + 28 \cdot t = \gcd(161, 28)$$

q	r_1	r_2	r	s_1	s_2	s	t_1	t_2	t
5	161	28	21	1	0	1	0	1	-5
1	28	21	7	0	1	-1	1	-5	6
3	21	7	0	1	-1	4	-5	6	-23
	7	0		-1	4		6	-23	

- Hence, we have $\gcd(161, 28) = 7$, $s = -1$ and $t = 6$ such that

$$161(-1) + 28(6) = 7$$

- **Example 2:** Find the $\gcd(97,25)$. Next, find the values of s and t such that

$$97 \cdot s + 25 \cdot t = \gcd(97, 25)$$

[illegible]

Divisibility – The Extended Euclidean Algorithm

- Example 2: Find the $\gcd(97,25)$. Next, find the values of s and t such that

$$97 \cdot s + 25 \cdot t = \gcd(97,25)$$

q	r_1	r_2	r	s_1	s_2	s	t_1	t_2	t
3	97	25	22	1	0	1	0	1	-3
1	25	22	3	0	1	-1	1	-3	4
7	22	3	1	1	-1	8	-3	4	-31
3	3	1	0						

- Hence, we have $\gcd(97,25) = 1$, $s = 8$ and $t = -31$ such that

$$97(\mathbf{8}) + 25(\mathbf{-31}) = \mathbf{1}$$

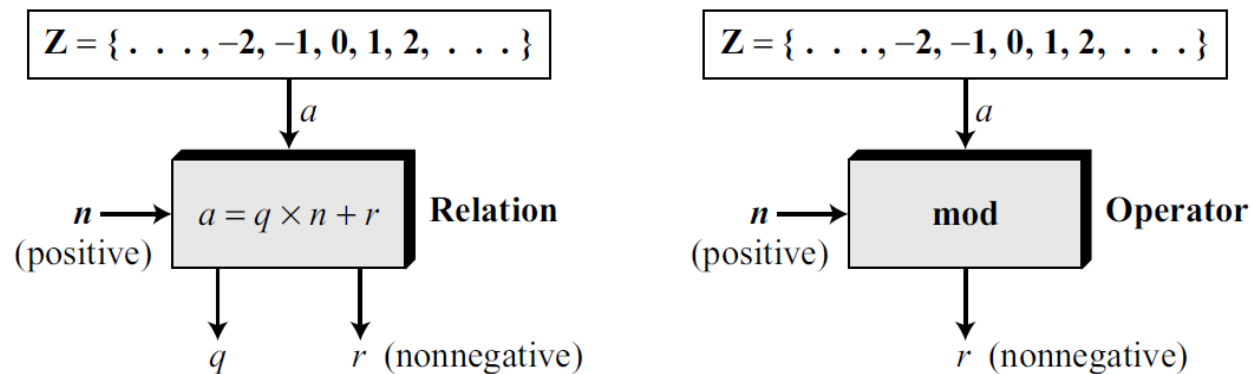
MODULAR ARITHMETIC



Modular Operator

- From **division relation** $a = q \cdot n + r$, we are now interested only on the **remainder** r when a is divided by n . This resembles the idea of modular arithmetic.
- The modular operator, or simply **mod** when expressing such an arithmetic, and the remainder is often called a **residue** r .

Figure 2.9 *Division relation and modulo operator*



Modular Operator

- **Examples:**

1. $27 \bmod 5 = 2$ since $27 = 5(5) + 2$.
2. $36 \bmod 12 = 0$ since $36 = 12(3) + 0$.
3. $-18 \bmod 14 = 10$ since $-18 = 14(-2) + 10$.
4. $-7 \bmod 10 = 3$ since $-7 = 10(-1) + 3$.

- Remember, when handling **negative numbers**, use the '**trade in**' concept to return **remainder r** into a **positive integer**.

Modular Arithmetic – Set of Residues

- Definition 3:

A set of least residues modulo n , or $\mathbb{Z}_n$ is the set of Integers between 0 and $n - 1$.

- Examples:

1. $\mathbb{Z}_2 = \{0,1\}$

2. $\mathbb{Z}_7 = \{0,1,2,3,4,5,6\}$

3. $\mathbb{Z}_{12} = \{0,1,2,3,4,5,6,7,8,9,10,11\}$

Modular Arithmetic – Congruence

- In cryptography, often the modular arithmetic is expressed in the concept of **congruence** (the congruence operator ' $\equiv$ ') instead of equality. Such a relation can be defined easily using a **notation ($\bmod n$) at the end of the expression**.
- Examples:
 1. $5 \equiv 17 \pmod{12} \equiv 29 \pmod{12} \equiv 41 \pmod{12}$
 2. $23 \equiv 33 \pmod{5} \equiv 38 \pmod{5} \equiv 48 \pmod{5}$
 3. $-18 \equiv -8 \pmod{10} \equiv 2 \pmod{10} \equiv 32 \pmod{10}$
 4. $0 \equiv 7 \pmod{7} \equiv 21 \pmod{7} \equiv 49 \pmod{7} \equiv 343 \pmod{7}$

Modular Arithmetic – Congruence and Residue

- Although one can cycle the modular operator endlessly, we are interested only in the relationship between the congruent modulo n with the set of residue $\mathbb{Z}_n$.
- In other words, we are interested in finding the **least positive integer modulo n** that is in the set of $\mathbb{Z}_n$

Modular Arithmetic

▪ **Examples:** Perform the following modular arithmetic

1. $(14 + 7) \pmod{15} \equiv 21 \pmod{15} \equiv 6$

2. $(3 - 14) \pmod{8} \equiv -11 \pmod{8} \equiv 5$

3. $(7 \times 5) \pmod{13} \equiv 35 \pmod{13} \equiv 9$

4. $(3 + 4 \times 7) \pmod{20} \equiv 31 \pmod{20} \equiv 11$

Modular Arithmetic – Properties

■ Given any integers a, b and a modular n :

1. $(a + b)(\text{mod } n) \equiv [(a \text{ mod } n) + (b \text{ mod } n)] (\text{mod } n)$

2. $(a - b)(\text{mod } n) \equiv [(a \text{ mod } n) - (b \text{ mod } n)] (\text{mod } n)$

3. $(a \times b)(\text{mod } n) \equiv [(a \text{ mod } n) \times (b \text{ mod } n)] (\text{mod } n)$

❖ In other words, we can perform modular operator first, followed by binary operation, OR we can perform binary operation first, followed by modular operation. The order is NOT important.

Modular Arithmetic – Properties

- Examples:

1. $(35 + 74)(\text{mod } 12) \equiv (11 + 2)(\text{mod } 12) \equiv 13(\text{mod } 12) \equiv 1$

2. $(35 + 74)(\text{mod } 12) \equiv 109(\text{mod } 12) \equiv 1$

3. $(24 - 33)(\text{mod } 8) \equiv (0 + 7)(\text{mod } 8) \equiv 7$

4. $(24 - 33)(\text{mod } 8) \equiv -9(\text{mod } 8) \equiv 7$

5. $(13 \times 71)(\text{mod } 23) \equiv (13 \times 2)(\text{mod } 23) \equiv 26(\text{mod } 23) \equiv 3$

6. $(13 \times 71)(\text{mod } 23) \equiv 923(\text{mod } 23) \equiv 3$

Did you notice the **difference** in each pair of examples?

Modular Arithmetic – Inverses

- In modular arithmetic, all the integers in a group $\mathbb{Z}_n$ are the set of least residue modulo n . These residue will cycle over and over again in binary operations.
- Hence, the following **two inverses** are defined in the modular arithmetic:
 1. **Additive inverse.**
 2. **Multiplicative inverse.**

Modular Arithmetic – Additive Inverses

- Definition 4:

In $\mathbb{Z}_n$, two numbers a and b are additive inverses of each other if

$$a + b \equiv 0 \pmod{n}$$

- Each integer in modular arithmetic has an additive inverse, it can found simply by performing subtraction in the normal fashion.

Modular Arithmetic – Additive Inverses

- Examples:

1. In $\mathbb{Z}_7$, 1 and 6 are additive inverses of each other since $1 + 6 \equiv 7 \equiv 0 \pmod{7}$.
 2. In $\mathbb{Z}_{21}$, the additive inverse of 9 is $21 - 9 \equiv 12 \pmod{21}$.
 3. In $\mathbb{Z}_{11}$, the integer pairs $(0,0)$, $(1,10)$, $(2,9)$, $(3,8)$, $(4,7)$, $(5,6)$ are additive inverses to each other.
- ❖ Can you build your own group of integer $\mathbb{Z}_n$ and find their corresponding additive inverses?

Modular Arithmetic – Multiplicative Inverses

- Definition 5:

In $\mathbb{Z}_n$, two numbers a and b are multiplicative inverses of each other if

$$a \times b \equiv 1 \pmod{n}$$

- In modular arithmetic, **NOT** every integer has a multiplicative inverse. The multiplicative inverse of an integer a in $\mathbb{Z}_n$ exists **IF AND ONLY IF** $\gcd(a, n) = 1$, that is the integers are **RELATIVELY PRIME**.

Modular Arithmetic – Multiplicative Inverses

■ Examples:

1. In $\mathbb{Z}_7$, the multiplicative inverse of 5 is 3 since $5 \times 3 \equiv 15 \equiv 1 \pmod{7}$.
 2. In $\mathbb{Z}_{15}$, the multiplicative inverse of 11 is 11 since $11 \times 11 \equiv 121 \equiv 1 \pmod{15}$.
 3. In $\mathbb{Z}_9$, the multiplicative inverse of 6 **DOES NOT** exist since $\gcd(6,9) = 3 \neq 1$.
 4. In $\mathbb{Z}_{11}$, the integer pairs (1,1), (2,6), (3,4), (5,9), (7,8), (10,10) are multiplicative inverses of each other.
- ❖ Can you build your own group of integer $\mathbb{Z}_n$ and find their corresponding multiplicative inverses?

Modular Arithmetic – Multiplicative Inverses

- Finding multiplicative inverse: The Extended Euclidean Algorithm.

➤ We learned in Definition 2 that:

Given two integers a and n , there exist another two integers s and t such that

$$s \cdot a + t \cdot n = \gcd(a, n)$$

➤ If the $\gcd(a, n) = 1$, then the integer s is the multiplicative inverse of a .

Modular Arithmetic – Multiplicative Inverses

▪ **Example:** Find the multiplicative inverse (if exists) of 25 in $\mathbb{Z}_{97}$.

➤ **Solution:** By the Extended Euclidean Algorithm.

q	r_1	r_2	r	s_1	s_2	s	t_1	t_2	t
3	97	25	22	1	0	1	0	1	-3
1	25	22	3	0	1	-1	1	-3	4
7	22	3	1	1	-1	8	-3	4	-31
3	3	1	0						

Modular Arithmetic – Multiplicative Inverses

▪ **Example:** Find the multiplicative inverse (if exists) of 25 in $\mathbb{Z}_{97}$.

➤ **Solution:** By the Extended Euclidean Algorithm.

q	r_1	r_2	r	s_1	s_2	s	t_1	t_2	t
3	97	25	22	1	0	1	0	1	-3
1	25	22	3	0	1	-1	1	-3	4
7	22	3	1	1	-1	8	-3	4	-31
3	3	1	0						

➤ Since the $\gcd(25, 97) = 1$. Hence the **multiplicative inverse of 25 exists** and is $-31 \equiv \mathbf{66} \pmod{97}$.

-

Modular Arithmetic – Multiplicative Inverses

■ **Exercise:** Find the multiplicative inverse (if exists) of 13 in $\mathbb{Z}_{23}$.

➤ **Solution:** By the Extended Euclidean Algorithm.

q	r_1	r_2	r	s_1	s_2	s	t_1	t_2	t

➤ Since the $\gcd(13, 23) = 1$. Hence the **multiplicative inverse of 13 exists** and is $-7 \equiv \mathbf{16} \pmod{23}$.

MATRICES

(Optional)



Matrices

- Definition 6:

*A matrix is a rectangular array with $(l \times m)$ elements, in which l indicates the number of **ROWS** and m indicates the number of **COLUMNS**.*

$$\begin{array}{c} l \text{ rows} \end{array} \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \vdots & \vdots & & \vdots \\ a_{l1} & a_{l2} & \dots & a_{lm} \end{bmatrix} \begin{array}{c} m \text{ columns} \end{array}$$

In general, $(l \times m)$ is called the **dimension** of the matrix

Matrices – Types of Matrices

- **Identity Matrix:** $I_{2 \times 2} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $I_{3 \times 3} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. The identity matrix defined **only** over square matrix.
- **Square Matrix:** Matrix with equal numbers of row and column.
- **Row Matrix:** $[1 \quad 4 \quad 6]$.
- **Column Matrix:** $\begin{bmatrix} 6 \\ 7 \\ 3 \end{bmatrix}$

Matrices – Operations and Relations

- In linear algebra, several operations and relations are defined over matrices.
- 1. **Equality**: Two matrices A and B are **equal** if they have the **same dimension** and all the **elements are the same**. In other words, $A = B$ if $a_{ij} = b_{ij}$ for all i and j .
- 2. **Addition and Subtraction**: Two matrices A and B can be added and subtracted in the **same dimension**. If $C = A + B$, then $c_{ij} = a_{ij} + b_{ij}$. Similarly, if $C = A - B$, then $c_{ij} = a_{ij} - b_{ij}$.

Matrices – Operations and Relations

3. **Multiplication**: Two matrices A and B can be multiplied only if the column of the matrix A is the same as the row of the matrix B . That is, $(l \times m)_A$ and $(m \times n)_B$. Mathematically, if $C = A \times B$, then $c_{ik} = \sum a_{ij} \times b_{jk}$.
4. **Scalar Multiplication**: Any matrix can be multiplied to a constant number (scalar). Mathematically, if $C = kA$, then $c_{ij} = k(a_{ij})$.

Matrices – Operations and Relations

- Examples:

1. $[1 \ 4 \ 6] + [6 \ 8 \ 7] = [7 \ 12 \ 13].$

2. $\begin{bmatrix} 6 \\ 7 \\ 3 \end{bmatrix} - \begin{bmatrix} 4 \\ 9 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \\ 3 \end{bmatrix}.$

3. $\begin{bmatrix} 1 & 3 & 2 \\ 7 & 4 & 1 \end{bmatrix} \times \begin{bmatrix} 2 & 4 \\ 7 & 6 \\ 5 & 1 \end{bmatrix} \begin{matrix} \xrightarrow{\text{red arrow}} \\ \downarrow \text{red arrow} \end{matrix} = \begin{bmatrix} 2 + 21 + 10 & 4 + 18 + 2 \\ 14 + 27 + 5 & 28 + 24 + 1 \end{bmatrix} = \begin{bmatrix} 33 & 24 \\ 46 & 53 \end{bmatrix}$

4. $5[6 \ 8 \ 7] = [30 \ 40 \ 35]$

Matrices – Determinant and Inverses

- Like integers, matrices have both **additive** and **multiplicative inverses**.
- **Additive Inverse**: Two matrices A and B are additive inverses to each other if $A + B = 0$. In other words, $A = -B$.
- **Multiplicative Inverse**. Two matrices A and B are multiplicative inverses to each other if $A \times B = I$. In other words, $A = B^{-1}$. **Multiplicative inverse defined only for square matrix**, with the condition that its **determinant, $\det(A) \neq 0$** .
- ❖ Please refer to any **linear algebra reference** for a complete demonstrations on matrices' operations.

Matrices

- In this course, we **do not** cover much about Matrices, since the only function of this concept is to learn how to operate the classical cipher in the next chapter.

Main References for Chapter 2

1. Forouzan, B.A. “*Introduction to Cryptography and Network Security*”, Third Edition, McGraw-Hill, 2008.

~ THE END 😊 ~